

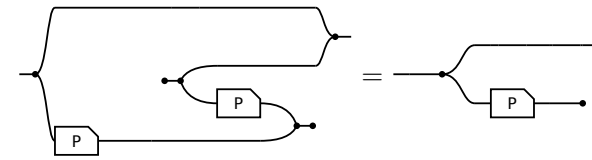
Relation Algebra

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

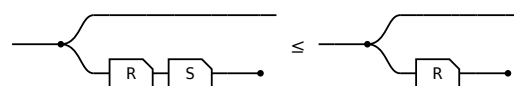
(5c)

1	$\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$
2	$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X \circ R$
3	$\text{dom}(R \circ S) \sqsubseteq \text{dom}(R)$
4	$\text{dom}(R \sqcap S) = \mathbb{1} \sqcap (S \circ R^\circ)$
5	$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$
6	$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ \circ R \sqsubseteq \mathbb{1}$
7	$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
8	$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \circ S)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \circ g)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \circ g)$
9	$\text{Entire}(R \circ S) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \sqsubseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

Equality is S entire, which is the same picture read as $\text{dom}(R) = \mathbb{1} \iff R \text{ entire}$.